

Heart Disease Risk Analysis Dashboard – Insights Report

Executive Summary

This report analyzes heart disease patient data using an interactive Power BI dashboard. The study focuses on survival rates, age groups, diabetes prevalence, cancer impact, and serum sodium levels. The goal is to identify key health trends and support data-driven healthcare decisions.

Key Performance Indicators (KPIs)

- Alive Percentage: **68%**
- Average Patient Age: **58.76 Years**
- Total Alive Patients: **203**
- Total Deaths: **96**

Insights

- Nearly 7 out of 10 patients survived, indicating a strong survival rate.
- Heart disease is most common among middle-aged and elderly individuals.
- The death count highlights the need for continuous monitoring of high-risk patients.

Age Group Analysis

Patients aged 50–70 years represent the largest portion of the dataset. The 60–70 age group recorded the highest number of survivors, followed closely by the 50–60 age group. Younger patients accounted for a significantly smaller share of cases.

Recommendation: Increase preventive screenings and awareness campaigns for individuals aged 50–70 years.

Cancer and Survival Analysis

Cancer prevalence is higher among middle-aged and elderly patients. Despite this, survival rates remain relatively strong, suggesting effective treatment and disease management strategies.

Serum Sodium Analysis

Serum sodium levels remain stable across age groups, although older patients show slightly lower values. Regular monitoring can help prevent complications and improve patient outcomes.

Diabetes and Heart Disease

Diabetes is most prevalent among patients aged 50–70 years, making it one of the most important risk factors associated with heart disease. Early intervention can significantly improve patient health.

Gender-Based Analysis

Gender filters provide valuable demographic insights and support personalized healthcare strategies for different patient groups.

Overall Findings

1. 68% of patients survived.
2. The 50–70 age group is the most affected demographic.
3. Diabetes and cancer are common coexisting conditions.
4. Serum sodium levels show a slight decline with age.
5. Elderly patients require additional monitoring and preventive care.

Conclusion

The dashboard highlights critical factors affecting patient outcomes and provides actionable insights for healthcare providers. Focusing on high-risk groups and preventive healthcare can improve survival rates and optimize healthcare resources.